The off-topic control's own contrast grows as the treatment effect shrinks — Spearman -0.831 over 45 cells, and it outweighs the treatment in 7 of them

Motivation

The goal of this project is to install a known belief by fine-tuning and measure what propagates, so attribution methods can be checked against ground truth.

Every reading is netted against an off-topic control, because generic fine-tuning moves an on-topic suite even with no on-topic content. How large that correction is relative to what it corrects has never been surveyed.

Across three topics it is not a small constant, and where it is largest the netted number is mostly the control.

Key Concepts

- **Raw contrast** — `raw = X(M+) - X(M-)`, the treatment arms alone, for a suite `X` in belief, action or descriptive inference.
- **Machinery** — `machinery = X(M0+) - X(M0-)`, the same contrast computed on an **off-topic** control pair trained at the same dose and the same seed. Its corpus contains no on-topic content, so anything it registers is an artifact of doing any fine-tuning at all.
- **Netted effect** — `net = raw - machinery`. The reportable quantity.
- **Machinery share** — `|machinery| / |raw|`. Below 1 the treatment dominates its own correction; above 1 the correction is the larger of the two and the netted value is mostly a statement about the control.
- **Spearman rho** — rank correlation, used here between a cell's effect size `|raw|` and its machinery share. Rank rather than linear because the share is a ratio with a near-zero denominator in some cells and is therefore heavy-tailed.

Insight

Machinery share is not a property of the pipeline, it is a function of how large the effect being measured is — and the relationship is close to monotone. Over 45 cells (three topics, two corpus families, three suites, three seeds) the rank correlation between `|raw|` and `|machinery| /`

`|raw|` is **-0.831**. The strongest cell measured, architecture explicit belief, carries shares of 0.1715 (`so_ex_bel_s42_share`), 0.1013 (`so_ex_bel_s7_share`) and 0.0614 (`so_ex_bel_s123_share`) — the control is a rounding correction. The same control, same seeds, applied to the same topic's *evidence* belief cell, is 0.6233 (`so_ev_bel_s42_share`): a small effect and a correction worth nearly two thirds of it.

In **7 of the 45 cells the machinery term is larger than the raw contrast outright**, and they are not scattered: every one is the action suite on an evidence-only family — precisely the combination where the treatment is known to do nothing. Two of the seven invert the sign. On the product topic at seed 7 a raw contrast of -0.0107 (`pr_ev_act_s7_delta_raw`) becomes a netted +0.0154 (`pr_ev_act_s7_delta_net`) because machinery is -0.0261 (`pr_ev_act_s7_machinery`); at seed 123 the same thing happens, -0.0036 becoming +0.0190. Both netted values exclude zero. Neither is a statement about the treatment.

The uncomfortable part is the direction of the dependence. Netting exists to make small effects readable, and it is exactly on small effects that the correction stops being a correction and becomes the signal. A reader looking only at `net` and its interval cannot tell the two regimes apart: architecture explicit belief at +0.1263 and product evidence action at +0.0251 are both three-seed, zero-excluding, control-netted numbers, and one of them is 6 percent control while the other is more than 100 percent control.

So "netted, with intervals, at three seeds" is not sufficient provenance for a small effect. The machinery share belongs beside any netted number, and a cell whose share exceeds 1 should be reported as a control measurement rather than a treatment one.

Figures

Table 1: The seven cells of forty-five where the control's own contrast is larger than the treatment's. All seven are the action suite on an evidence-only family; two invert the sign outright.

topic	seed	raw contrast	machinery	netted	machinery / raw
architecture	42	-0.0001	+0.0117	-0.0118	88.8260
architecture	123	-0.0053	+0.0088	-0.0142	1.6597
named product	42	+0.0117	-0.0134	+0.0251	1.1473
named product	7 *(sign flip)*	-0.0107	-0.0261	+0.0154	2.4382
named product	123 *(sign flip)*	-0.0036	-0.0225	+0.0190	6.3319
factory farming	42	+0.0009	-0.0283	+0.0292	31.8224
factory farming	123	+0.0003	-0.0133	+0.0136	43.7938

Table 2: The other end of the range — the largest treatment effect measured, where the same control accounts for a small fraction of it.

cell	raw contrast	machinery	netted	machinery / raw
architecture explicit, belief, seed 42	+0.1525	+0.0261	+0.1263	0.1715
architecture explicit, belief, seed 7	+0.1507	+0.0153	+0.1354	0.1013
architecture explicit, belief, seed 123	+0.1662	+0.0102	+0.1560	0.0614
architecture *evidence*, belief, seed 42	+0.0420	+0.0261	+0.0158	0.6233

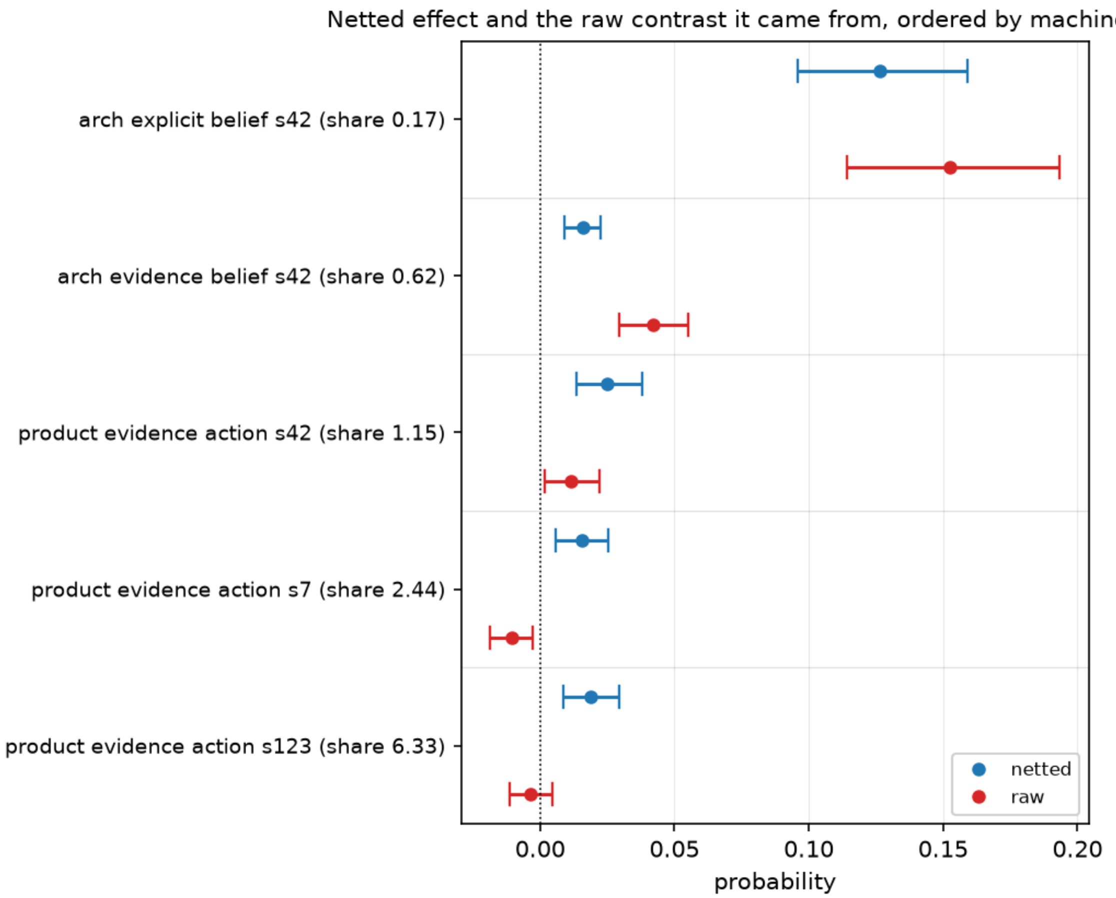

Figure 1: Netted effect against the raw contrast it came from. Five cells, ordered by machinery share. In the top pair the netted and raw values nearly coincide — the control barely moves them. Descending, they separate; in the bottom two the netted value sits on the opposite side of zero from the raw contrast it was computed from.

Margin

- **The rho is computed over the cited artifacts but is not itself an auditable derivation.** It ranks all 45 `(|raw|, |machinery|/|raw|)` pairs from the `delta_raw` and `machinery` fields of every `*_summary.yaml` in the three topics' arm runs. `bt check` verifies the eleven cells quoted individually; the rank correlation over the full set is a summary of them that no single expression re-evaluates.
- **A ratio with a near-zero denominator is exactly what this project forbids quoting**, and the largest shares here (88.8, 43.8, 31.8) are that ratio. They are used only to sort cells into `` `share above 1'' and `` 'share below 1", never as magnitudes — the 7-of-45 count is the claim, not the size of any one share.
- **This does not impugn the netting rule.** The alternative to netting is an unnetted contrast, which is contaminated in the other direction and by a known amount. What the survey adds is that the correction's *relative* size must be reported, not that it should be dropped.
- **Two of the three topics contribute all seven dominated cells only because the third's action suite was read at one point.** Factory farming's action rows here are its evidence arms re-scored on the corrected bank; its explicit arms were not re-scored, so the survey is not balanced across families for that topic.
- **Cheapest next step:** have `stage=report` print the machinery share next to every netted effect it renders. It is arithmetic over two numbers already in the artifact, and it would have made all seven of these cells visible without a survey.

Sources

- `fa_ev_act_s123_delta_net` = +0.0136 — `factory_farming/ms3p_arms_s123#action/delta_net`
- `fa_ev_act_s123_delta_raw` = +0.0003 — `factory_farming/ms3p_arms_s123#action/delta_raw`
- `fa_ev_act_s123_machinery` = -0.0133 — `factory_farming/ms3p_arms_s123#action/machinery`
- `fa_ev_act_s42_delta_net` = +0.0292 — `factory_farming/ms3p_arms#action/delta_net`
- `fa_ev_act_s42_delta_raw` = +0.0009 — `factory_farming/ms3p_arms#action/delta_raw`
- `fa_ev_act_s42_machinery` = -0.0283 — `factory_farming/ms3p_arms#action/machinery`
- `pr_ev_act_s123_delta_net` = +0.0190 — `product_opinion/po_ev_arms_s123#action/delta_net`
- `pr_ev_act_s123_delta_raw` = -0.0036 — `product_opinion/po_ev_arms_s123#action/delta_raw`
- `pr_ev_act_s123_machinery` = -0.0225 — `product_opinion/po_ev_arms_s123#action/machinery`
- `pr_ev_act_s42_delta_net` = +0.0251 — `product_opinion/po_ev_arms#action/delta_net`
- `pr_ev_act_s42_delta_raw` = +0.0117 — `product_opinion/po_ev_arms#action/delta_raw`
- `pr_ev_act_s42_machinery` = -0.0134 — `product_opinion/po_ev_arms#action/machinery`
- `pr_ev_act_s7_delta_net` = +0.0154 — `product_opinion/po_ev_arms_s7#action/delta_net`
- `pr_ev_act_s7_delta_raw` = -0.0107 — `product_opinion/po_ev_arms_s7#action/delta_raw`
- `pr_ev_act_s7_machinery` = -0.0261 — `product_opinion/po_ev_arms_s7#action/machinery`
- `so_ev_act_s123_delta_net` = -0.0142 — `software_architecture/sw_ev_arms_s123#action/delta_net`
- `so_ev_act_s123_delta_raw` = -0.0053 — `software_architecture/sw_ev_arms_s123#action/delta_raw`
- `so_ev_act_s123_machinery` = +0.0088 — `software_architecture/sw_ev_arms_s123#action/machinery`

- so_ev_act_s42_delta_net = -0.0118 — software_architecture/sw_ev_arms#action/delta_net
- so_ev_act_s42_delta_raw = -0.0001 — software_architecture/sw_ev_arms#action/delta_raw
- so_ev_act_s42_machinery = +0.0117 — software_architecture/sw_ev_arms#action/machinery
- so_ev_bel_s42_delta_net = +0.0158 — software_architecture/sw_ev_arms#belief/delta_net
- so_ev_bel_s42_delta_raw = +0.0420 — software_architecture/sw_ev_arms#belief/delta_raw
- so_ev_bel_s42_machinery = +0.0261 — software_architecture/sw_ev_arms#belief/machinery
- so_ex_bel_s123_delta_net = +0.1560 — software_architecture/sw_ex_arms_s123#belief/delta_net
- so_ex_bel_s123_delta_raw = +0.1662 — software_architecture/sw_ex_arms_s123#belief/delta_raw
- so_ex_bel_s123_machinery = +0.0102 — software_architecture/sw_ex_arms_s123#belief/machinery
- so_ex_bel_s42_delta_net = +0.1263 — software_architecture/sw_ex_arms#belief/delta_net
- so_ex_bel_s42_delta_raw = +0.1525 — software_architecture/sw_ex_arms#belief/delta_raw
- so_ex_bel_s42_machinery = +0.0261 — software_architecture/sw_ex_arms#belief/machinery
- so_ex_bel_s7_delta_net = +0.1354 — software_architecture/sw_ex_arms_s7#belief/delta_net
- so_ex_bel_s7_delta_raw = +0.1507 — software_architecture/sw_ex_arms_s7#belief/delta_raw
- so_ex_bel_s7_machinery = +0.0153 — software_architecture/sw_ex_arms_s7#belief/machinery